

CSA07 – Computer Networks (CO5)

Answered Pseudocode Writing Task – Algorithm Design

The answers below arrange all provided pseudocode blocks into a valid algorithm sequence, without adding new blocks.

Question 1: P2P File Sharing Algorithm

Topic: P2P Communication

Correct Algorithm:

```
START
CONNECT TO PEER
CHECK WHETHER THE PEER HAS THE REQUESTED FILE
IF FILE IS AVAILABLE THEN
    DOWNLOAD FILE
ELSE
    DISPLAY "File not available"
ENDIF
END
```

Explanation: The process starts by connecting to a peer. The system then checks for the requested file. If the file exists, it is downloaded; otherwise, the failure message is displayed.

Question 2: VoIP Call Connection Algorithm

Topic: VoIP Communication

Correct Algorithm:

```
START
CHECK INTERNET CONNECTION
IF INTERNET IS AVAILABLE THEN
    CONNECT CALL
    START VOICE COMMUNICATION
ELSE
    DISPLAY "No Internet Connection"
ENDIF
END
```

Explanation: The application first checks the internet connection. If the connection is available, it connects the call and starts voice communication. Otherwise, it displays the specified error message.

Question 3: Overlay Network Message Routing Algorithm

Topic: Overlay Networks

Correct Algorithm:

```
START
CHECK DESTINATION NODE
IF DESTINATION NODE IS AVAILABLE THEN
    SEND MESSAGE DIRECTLY
ELSE
    FORWARD MESSAGE THROUGH ANOTHER AVAILABLE NODE
ENDIF
DISPLAY "MESSAGE SENT"
END
```

Explanation: The destination node is checked first. If it is available, the message is sent directly. If it is unavailable, the message is forwarded through another available node. The algorithm then confirms that the message has been sent and ends.

Note: These sequences are based directly on the blocks and constraints shown in the uploaded question sheet.